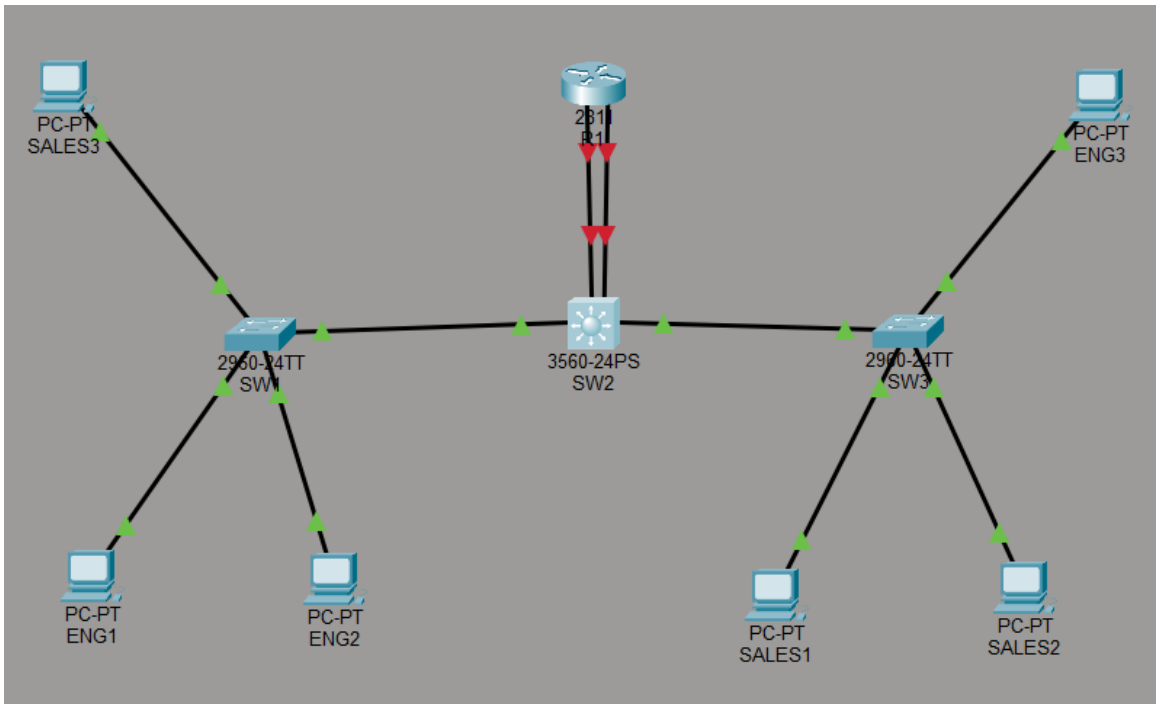
Network Topology



VTP Configuration

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastethernet0/24
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet0/23
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#end
Switch#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                 Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22
10   ENG                               active    Fa0/24
20   SALES                             active    Fa0/23
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

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```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#interface fastethernet0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.1      YES manual up                    up
FastEthernet0/1        10.10.20.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

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SW3: show vtp status

```
Pinging 10.10.20.11 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time=1ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

VLAN Configuration Screenshot 1

```
Building configuration...
[OK]
Switch#
Switch#enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1
10   ENG                              active    Fa0/1
20   SALES                            active    Fa0/2, Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

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VLAN Configuration Screenshot 2

```
Building configuration...
[OK]
Switch#
Switch#enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1
10   ENG                              active    Fa0/1
20   SALES                            active    Fa0/2, Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/2      10,20

Port        Vlans allowed and active in management domain
Gig0/2      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/2      10,20

Switch#
```

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SW1 and SW3 VLAN screenshots

```
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      10,20
Gig0/1      10,20
Gig0/2      10,20

Port        Vlans allowed and active in management domain
Fa0/24      10,20
Gig0/1      10,20
Gig0/2      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      none
Gig0/1      10,20
Gig0/2      10,20

Switch#write memory
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

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Trunk Screenshot 1

```
Building configuration...
[OK]
Switch#
Switch#enable
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20
Gig0/2      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20
Gig0/2      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20
Gig0/2      10,20

Switch#
```

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Trunk Screenshot 2

```
Switch>enable
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20

Switch#
```

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```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES manual up                    up
FastEthernet0/0.10     10.10.10.1      YES manual up                    up
FastEthernet0/0.20     10.10.20.1      YES manual up                    up
FastEthernet0/1        unassigned      YES manual administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#write memory
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

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Task 93 — Option 1: Separate Router Interfaces

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 20
Switch(config-if)#ip address 10.10.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down

Switch#rite memory
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

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Inter-VLAN Ping

```
Pinging 10.10.20.11 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time=1ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Task 94 — Option 2: Router-on-a-Stick

```
C:\>ping 10.10.20.11

Pinging 10.10.20.11 with 32 bytes of data:

Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 95 — Option 3: Layer 3 Switch

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 20
Switch(config-if)#ip address 10.10.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down

Switch#rite memory
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

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Task 96 — Reflection

A Layer 3 switch is the best option for a large campus network because it can route traffic between VLANs faster and more efficiently, it can handle many VLANs without needing separate router connections, it is also easier to manage and expand when more devices are added to the network.